

The extremal problems on the terminal Mostar index of unicyclic graphs with prescribed cycle length and diameter*

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Abstract

The Mostar index is a bond-additive distance-based topological index that measures the distance imbalance of the two end-vertices of an edge. Motivated by the terminal Mostar index of trees, we extend the definition to connected graphs by counting pendent vertices only. For an edge uv of a connected graph G , let $l_u(uv)$ be the number of leaves closer to u than to v , and define

$$\text{Mo}_t(G) = \sum_{uv \in E(G)} |l_u(uv) - l_v(uv)|.$$

We study this invariant on unicyclic graphs. First, we derive basic contribution formulae for cut edges and cycle edges. Then we determine the minimum and maximum values among all unicyclic graphs of fixed order and among all unicyclic graphs with prescribed cycle length. We further consider the class with both prescribed cycle length and prescribed diameter: a closed formula for the maximum is obtained, together with the corresponding extremal structures. For the minimum, we characterize the zero cases, the odd-cycle maximal-diameter critical case, the two-leaf subfamily, and an exact finite balanced optimization formulation for the remaining cases. As a consequence, the maximum for unicyclic graphs with prescribed diameter is also expressed explicitly. The results show that the terminal Mostar index has extremal graphs different from those of the ordinary Mostar index.

Keywords: Mostar index; terminal Mostar index; unicyclic graph; cycle length; diameter; extremal graph

1 Introduction

Distance-based topological indices form an important family of invariants in chemical graph theory. The Mostar index was introduced by Došlić et al. as an edge-additive measure of distance imbalance. For an edge uv of a connected graph G , let $n_u(uv)$ be the number of vertices closer to u than to v . The ordinary Mostar index is

$$\text{Mo}(G) = \sum_{uv \in E(G)} |n_u(uv) - n_v(uv)|.$$

Basic properties, bounds and extremal problems for the Mostar index are surveyed in [1].

The same survey also discusses the terminal Mostar index of a tree T , where only leaves are counted:

$$\text{Mo}_t(T) = \sum_{uv \in E(T)} |l_u(uv) - l_v(uv)|.$$

Here $l_u(uv)$ denotes the number of leaves closer to u than to v . For trees of order n , the path P_n and the star S_n are the unique extremal graphs for the minimum and maximum, respectively; with a fixed number of leaves, the balanced double broom and the simple broom are extremal [1].

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The purpose of this paper is to extend the terminal version naturally to graphs containing cycles and to study the first non-tree class, namely unicyclic graphs. Extremal unicyclic graphs for the ordinary Mostar index are known: the cycle C_n gives value zero, while the maximum is attained by S_n^+ or by a graph obtained from C_4 by attaching pendent vertices, depending on n [2, 1]. The terminal version behaves differently. For instance, for $n \geq 6$ the maximum terminal Mostar index among all n -vertex unicyclic graphs is attained uniquely by S_n^+ .

The paper is organized as follows. Section 2 contains definitions and elementary lemmas. Section 3 determines the extremal values for all unicyclic graphs of fixed order. Section 4 treats the class with prescribed cycle length. Section 5 studies the simultaneous restrictions of prescribed cycle length and prescribed diameter. Section 6 derives consequences for prescribed diameter alone, including the maximum value and the zero/minimum phenomena. Section 7 gives concluding remarks and further problems.

2 Preliminaries

All graphs considered in this paper are finite, simple and connected. For a graph G , let $V(G)$ and $E(G)$ denote its vertex set and edge set. The distance between vertices x and y is denoted by $d_G(x, y)$. A vertex of degree one is called a leaf or a pendent vertex. Let

$$L(G) = \{x \in V(G) : d_G(x) = 1\}, \quad k(G) = |L(G)|.$$

Definition 2.1. Let G be a connected graph and let $uv \in E(G)$. Define

$$l_u(uv) = |\{x \in L(G) : d_G(x, u) < d_G(x, v)\}|,$$

$$l_v(uv) = |\{x \in L(G) : d_G(x, v) < d_G(x, u)\}|.$$

If a leaf is equidistant from u and v , it is not counted for either end. The terminal Mostar index of G is

$$\text{Mo}_t(G) = \sum_{uv \in E(G)} |l_u(uv) - l_v(uv)|.$$

Let G be a unicyclic graph. Its unique cycle is denoted by $C(G)$, and its cycle length by $c(G)$. After deleting the cycle edges, G is a collection of rooted trees attached to the vertices of $C(G)$. The root on the cycle of the tree containing a leaf will be called the attachment vertex of that leaf. Put

$$\eta_c = \begin{cases} c, & c \text{ even,} \\ c - 1, & c \text{ odd.} \end{cases}$$

Lemma 2.2 (Contribution of cut edges). *Let G be a unicyclic graph and let $k = |L(G)|$. If $e = uv \notin E(C(G))$ is a cut edge and one component of $G - e$ contains s leaves, then the contribution of e to $\text{Mo}_t(G)$ is*

$$|k - 2s|.$$

In particular, if e is a leaf edge and $k \geq 2$, then its contribution is $k - 2$.

Proof. After deleting the cut edge e , every leaf lies in exactly one of the two components. Leaves on the u -side are closer to u than to v , and leaves on the v -side are closer to v than to u . If one side contains s leaves, the other contains $k - s$ leaves, so the contribution is $|s - (k - s)| = |k - 2s|$. The leaf-edge case corresponds to $s = 1$. $\square$

Lemma 2.3 (Upper bound for cycle-edge contributions). *Let G be a unicyclic graph with cycle length c and $k = |L(G)|$. Then the total contribution of all cycle edges is at most*

$$\eta_c k.$$

If $k > 0$ and all leaves are attached to the same cycle vertex, the bound is attained. Conversely, if the bound is attained, then all leaves are attached to one cycle vertex.

Proof. Fix a leaf x and let w be its attachment vertex on the cycle. For a cycle edge uv , the common distance from x to w cancels when comparing $d_G(x, u)$ and $d_G(x, v)$; hence only the distances in C_c from w to u and v matter. If c is even, these distances are different for every cycle edge; if c is odd, there is exactly one cycle edge whose two end-vertices are equidistant from w . Thus one leaf can contribute to at most c cycle edges in the even case and at most $c - 1$ in the odd case. Summing over leaves and using the triangle inequality gives the bound.

If all leaves are attached to one cycle vertex, all nonzero votes on each cycle edge have the same direction, so equality holds. Conversely, if two distinct attachment vertices occur, then on the first edge of a shortest arc between them the two groups of leaves vote in opposite directions, making the triangle inequality strict. Hence equality is impossible. $\square$

Lemma 2.4 (Two leaves on a cycle). *Assume that G has exactly two leaves, attached to two vertices of C_c whose distance on the cycle is t , where $0 \leq t \leq \lfloor c/2 \rfloor$. Then:*

- (i) *if $c = 2r$, the total cycle-edge contribution is $4(r - t)$; in particular it is zero if and only if $t = r$;*
- (ii) *if $c = 2r + 1$ and $t \geq 1$, the total cycle-edge contribution is $4(r - t) + 2$; in particular the minimum is 2, attained if and only if $t = r$.*

If c is odd and $t = 0$, the contribution is $2(c - 1) = 4r$.

Proof. For a fixed cycle edge, the two leaves either vote in the same direction, in opposite directions, or one leaf is equidistant from the two end-vertices. Only the same-direction case contributes 2. If the two attachment vertices split the cycle into arcs of lengths t and $c - t$, a direct count gives $2(r - t)$ same-direction edges for $c = 2r$, and $2(r - t) + 1$ same-direction edges for $c = 2r + 1$ with $t \geq 1$. This proves the stated formulae. The case $t = 0$ on an odd cycle has $c - 1 = 2r$ nonzero edges, each contributing 2. $\square$

3 Extremal values for fixed order

Let $\mathcal{U}_n$ be the set of all connected unicyclic graphs of order n .

Theorem 3.1 (Minimum). *For every $n \geq 3$,*

$$\min_{G \in \mathcal{U}_n} \text{Mo}_t(G) = 0.$$

Equality holds if and only if either $G \cong C_n$, or G is obtained from an even cycle C_{2r} by attaching two paths to two opposite cycle vertices. If the two path lengths are $p, q \geq 1$, then $2r + p + q = n$.

Proof. The cycle C_n has no leaves, so its terminal Mostar index is zero. In the second construction there are exactly two leaves. Each non-cycle edge has one leaf on each side and hence contributes zero; the two leaves are attached to opposite vertices of an even cycle, so the cycle contribution is also zero by the preceding lemma.

Conversely, suppose $\text{Mo}_t(G) = 0$ and G is not a cycle. Let $k = |L(G)|$. A leaf edge has contribution $|k - 2|$, so $k = 2$. Every non-cycle cut edge must then separate the two leaves; hence the non-cycle part consists of two pendent paths and no common stem or extra branch. The two-leaf cycle lemma forces the two attachment vertices to be opposite on an even cycle. $\square$

Theorem 3.2 (Maximum). *For $n \geq 3$,*

$$\max_{G \in \mathcal{U}_n} \text{Mo}_t(G) = \begin{cases} 0, & n = 3, \\ 3, & n = 4, \\ 5, & n = 5, \\ (n - 3)^2, & n \geq 6. \end{cases}$$

The extremal graph is C_3 for $n = 3$, S_4^+ for $n = 4$, the graph obtained from C_4 by attaching one leaf to a cycle vertex for $n = 5$, and uniquely S_n^+ for $n \geq 6$.

Proof. Let the unique cycle have length c , put $m = n - c$, and let k be the number of leaves. If $k = 0$, the graph is C_n and the value is zero. Assume $k \geq 2$. There are m non-cycle edges, among which k are leaf edges. The non-cycle contribution is at most $k(k - 2) + (m - k)k = k(m - 2)$. If c is even, the cycle contribution is at most ck , so

$$\text{Mo}_t(G) \leq k(n - 2) \leq (n - c)(n - 2) \leq (n - 4)(n - 2) < (n - 3)^2.$$

If c is odd, the cycle contribution is at most $(c - 1)k$, and

$$\text{Mo}_t(G) \leq k(n - 3) \leq (n - c)(n - 3) \leq (n - 3)^2.$$

Equality requires $c = 3$, $k = m = n - 3$, and all leaves attached to one cycle vertex, namely S_n^+ .

If $k = 1$, then the graph is a cycle with one pendent path. Its value is $m + \text{obreak}\eta_c$. A direct comparison for small n gives the exceptional cases $n = 4, 5$; for $n \geq 6$ this value does not exceed $(n - 3)^2$. $\square$

4 Extremal values for prescribed cycle length

Let

$$\mathcal{U}_{n,c} = \{G : G \text{ is a connected unicyclic graph of order } n \text{ and cycle length } c\},$$

where $3 \leq c \leq n$, and put $m = n - c$.

Theorem 4.1 (Minimum for prescribed cycle length). *For $3 \leq c \leq n$,*

$$\min_{G \in \mathcal{U}_{n,c}} \text{Mo}_t(G) = \begin{cases} 0, & m = 0, \\ 1 + \eta_c, & m = 1, \\ 0, & m \geq 2 \text{ and } c \text{ even}, \\ 2, & m \geq 2 \text{ and } c \text{ odd}. \end{cases}$$

The extremal graphs are as follows: C_c for $m = 0$; a single leaf attached to C_c for $m = 1$; two pendent paths attached to opposite vertices of C_c for even c and $m \geq 2$; and two pendent paths attached to two vertices at distance $(c - 1)/2$ for odd c and $m \geq 2$.

Proof. The cases $m = 0$ and $m = 1$ are immediate. Let $m \geq 2$. If G has one leaf, then it is a cycle with one pendent path and has value $m + \text{obreak}\eta_c$. If it has $k \geq 3$ leaves, the leaf edges alone contribute at least $k(k - 2) \geq 3$. Therefore the minimum is obtained in the two-leaf case. Then all non-cycle contributions can be made zero only when the two leaves lie at the ends of two separate pendent paths. The problem reduces to minimizing the cycle contribution, which is handled by the two-leaf cycle lemma. $\square$

Theorem 4.2 (Maximum for prescribed cycle length). *For $3 \leq c \leq n$,*

$$\max_{G \in \mathcal{U}_{n,c}} \text{Mo}_t(G) = \begin{cases} 0, & m = 0, \\ 1 + \eta_c, & m = 1, \\ m(m - 2 + \eta_c), & m \geq 2. \end{cases}$$

For $m \geq 2$, an extremal graph is obtained by attaching all m non-cycle vertices as leaves to one vertex of C_c . This extremal graph is unique except for $(c, m) = (3, 2)$, where the triangle with a pendent path of length two is also extremal.

Proof. The cases $m = 0, 1$ are immediate. Assume $m \geq 2$ and $k \geq 2$. The non-cycle contribution is at most $k(m - 2)$ and the cycle contribution at most $\eta_c k$, hence

$$\text{Mo}_t(G) \leq k(m - 2 + \eta_c) \leq m(m - 2 + \eta_c).$$

Equality forces $k = m$, so every non-cycle vertex is a leaf, and also forces equality in the cycle-edge bound, so all leaves are attached to one cycle vertex. If $k = 1$, the value is $m + \text{obreak}\eta_c$, which equals the above maximum only for $(c, m) = (3, 2)$. $\square$

5 Prescribed cycle length and prescribed diameter

We now study

$$\mathcal{U}_{n,c,d} = \{G : |V(G)| = n, G \text{ is unicyclic}, c(G) = c, \text{diam}(G) = d\}.$$

Let

$$m = n - c, \quad r = \left\lfloor \frac{c}{2} \right\rfloor, \quad \eta_c = \begin{cases} c, & c \text{ even}, \\ c - 1, & c \text{ odd}. \end{cases}$$

Proposition 5.1 (Feasible parameters). *The class $\mathcal{U}_{n,c,d}$ is nonempty if and only if either*

$$m = 0, \quad d = r,$$

or

$$m \geq 1, \quad r + 1 \leq d \leq r + m.$$

Proof. If $m = 0$, the only graph is C_c , whose diameter is r . Suppose $m \geq 1$. A leaf outside the cycle has distance at least $r + 1$ from a farthest cycle vertex, so $d \geq r + 1$. Any two vertices are at distance at most $m + r$, since at most all m non-cycle edges and a shortest cycle arc of length at most r are used; hence $d \leq r + m$. Conversely, for any d in this interval, attach a path of length $d - r$ to a cycle vertex and put all remaining non-cycle vertices, if any, as leaves at the same cycle vertex. The resulting graph has diameter d . $\square$

In the intermediate range $r + 2 \leq d \leq r + m - 1$, put

$$K = m - d + r + 2.$$

Then $K \geq 3$.

Theorem 5.2 (Maximum under both restrictions). *Assume $\mathcal{U}_{n,c,d}$ is nonempty. If $m = 0$, the maximum is 0. If $m \geq 1$ and $d = m + r$, then*

$$\max_{G \in \mathcal{U}_{n,c,d}} \text{Mo}_t(G) = m + \eta_c,$$

attained by a cycle with one pendent path of length m . If $m \geq 2$ and $d = r + 1$, then

$$\max_{G \in \mathcal{U}_{n,c,d}} \text{Mo}_t(G) = m(m - 2 + \eta_c),$$

attained by attaching m leaves to one vertex of C_c .

If $r + 2 \leq d \leq r + m - 1$, define

$$A = (K - 1)(m - 2 + \eta_c),$$

and

$$B = \begin{cases} n(K - 2), & c \text{ even}, \\ m(K - 2) + (c - 1)K - 2(c - 2), & c \text{ odd}. \end{cases}$$

Then

$$\max_{G \in \mathcal{U}_{n,c,d}} \text{Mo}_t(G) = \max\{A, B\}.$$

The value A is attained by a one-sided concentrated graph, and the value B by a two-sided strongly unbalanced graph. If $A = B$, both types are extremal.

Proof. The boundary cases follow from the definitions and from the fixed-cycle maximum theorem. Consider $r + 2 \leq d \leq r + m - 1$. Let k be the number of leaves and let $q = m - k$ be the number of non-leaf vertices outside the cycle. For $k \geq 2$, the non-cycle contribution is at most $k(m - 2)$ and the cycle contribution is at most $\eta_c k$, so

$$\text{Mo}_t(G) \leq k(m - 2 + \text{obreak} \eta_c). \quad (5.1)$$

Moreover, any leaf-to-leaf distance is at most $q + 2 + r$, and any leaf-to-cycle distance is at most $q + 1 + r$. Hence $d \leq q + 2 + r$, so

$$q \geq d - r - 2, \quad k \leq m - d + r + 2 = K. \quad (5.2)$$

If $k \leq K - 1$, (5.1) gives $\text{Mo}_t(G) \text{obreak} \leq A$. Equality is attained by taking a path of length $d - r - 1$ from a cycle vertex to a branching vertex and attaching $K - 1$ leaves there.

Suppose now $k = K$. Then equality holds in (5.2). A diametral leaf-to-leaf path must use all $q = d - r - 2$ non-leaf outside vertices and a cycle arc of length r . Hence all non-leaf outside vertices lie on two pendent channels whose roots on the cycle are at distance r . Every non-cycle edge then has contribution at most $K - 2$, so the non-cycle contribution is at most

$$m(K - 2). \quad (5.3)$$

If c is even, the two roots are opposite, and the maximum cycle contribution under two positive leaf groups occurs when the groups have sizes $K - 1$ and 1 . It is $c(K - 2)$, yielding $B = n(K - 2)$. If c is odd, the maximum cycle contribution for two positive groups at distance r is attained by the split $K - 1$ and 1 , and equals

$$(c - 1)K - 2(c - 2).$$

Together with (5.3), this gives the stated B . The two-sided construction with leaf-group sizes $K - 1$ and 1 and channel depths $a, b \geq 1$ satisfying $a + b = d - r$ attains the bound. $\square$

Remark 5.3. In the intermediate range, if c is even then $A = (K - 1)(n - 2)$ and $B = n(K - 2)$; hence $A \geq B$ if and only if $2K \leq n + 2$. If c is odd then $A = (K - 1)(n - 3)$ and $A - B = n - 1 - 2K$; hence $A \geq B$ if and only if $2K \leq n - 1$.

5.1 Minimum results

The maximum is governed by concentration of leaves. The minimum, by contrast, is governed by balancing leaf counts across cut edges and balancing directed leaf votes across cycle edges. With both c and d fixed, a single closed formula for all parameters is considerably more delicate. The following cases are exact.

Theorem 5.4 (Zero cases). *Assume $\mathcal{U}_{n,c,d}$ is nonempty. Then*

$$\min_{G \in \mathcal{U}_{n,c,d}} \text{Mo}_t(G) = 0$$

if and only if either

$$m = 0, \quad d = r,$$

or

$$c \text{ is even, } m \geq 2, \quad d = m + \frac{c}{2}.$$

In the latter case the extremal graphs are obtained from C_c by attaching two pendent paths of lengths $p, q \geq 1$, $p + q = m$, to two opposite cycle vertices.

Proof. The zero graphs among all unicyclic graphs are precisely cycles and even-cycle opposite double pendent paths. The first gives $m = 0, d = r$. In the second case the two leaves are at distance $p + q + c/2 = m + c/2$, which is the diameter. The converse is immediate from the construction. $\square$

Theorem 5.5 (Odd-cycle maximal-diameter critical case). *Let $c = 2r + 1$ be odd, $m \geq 2$, and $d = m + r$. Then*

$$\min_{G \in \mathcal{U}_{n,c,d}} \text{Mo}_t(G) = 2.$$

The extremal graphs are obtained by attaching two pendent paths of lengths $p, q \geq 1, p + q = m$, to two cycle vertices at distance r .

Proof. For the stated construction, all non-cycle edges contribute zero and the two-leaf cycle lemma gives cycle contribution 2. The zero theorem shows that value zero is impossible. A graph with one leaf has value $m + \text{obreak}\eta_c > 2$; a graph with at least three leaves has leaf-edge contribution at least 3. In the two-leaf case, any nonzero contribution is a nonnegative integer and the above construction attains the least positive possibility, namely 2. $\square$

Proposition 5.6 (The two-leaf subfamily). *Let $G \in \mathcal{U}_{n,c,d}$ have exactly two leaves and suppose all non-cycle vertices lie on two pendent paths. Let the path lengths be $p, q \geq 1$ with $p + q = m$, and let the distance between their attachment vertices on C_c be t , where $0 \leq t \leq r$. Then all non-cycle contributions are zero and*

$$\text{diam}(G) = \max\{m + t, \max\{p, q\} + r\}.$$

If $c = 2r$, then

$$\text{Mo}_t(G) = 4(r - t).$$

If $c = 2r + 1$, then

$$\text{Mo}_t(G) = \begin{cases} 4r, & t = 0, \\ 4(r - t) + 2, & 1 \leq t \leq r. \end{cases}$$

Consequently, within this two-leaf subfamily, the minimum for fixed n, c, d is obtained by maximizing t subject to

$$\max\{m + t, \max\{p, q\} + r\} = d.$$

Proof. Each pendent-path edge separates the two leaves and hence contributes zero. The diameter is the maximum of the distance between the two leaves, $p + q + t = m + t$, and the largest distance from a leaf to a cycle vertex, $\max\{p, q\} + r$. The stated formulae for $\text{Mo}_t(G)$ are exactly the two-leaf cycle lemma. $\square$

Proposition 5.7 (Finite optimization for the general minimum). *Fix $C_c = v_0 v_1 \cdots v_{c-1} v_0$. View a graph in $\mathcal{U}_{n,c,d}$ as a collection of rooted trees $(T_0, \dots, T_{c-1})$ attached to the cycle vertices and satisfying*

$$\sum_{i=0}^{c-1} (|V(T_i)| - 1) = m, \quad \text{diam}(G) = d.$$

Let $k = |L(G)|$. For a non-cycle edge e , let s_e be the number of leaves in the component of $G - e$ away from the cycle. For a cycle edge $e = v_i v_{i+1}$, let L_e^+ and L_e^- be the numbers of leaves closer to v_i and to v_{i+1} , respectively. Then

$$\text{Mo}_t(G) = \sum_{e \notin E(C_c)} |k - 2s_e| + \sum_{e \in E(C_c)} |L_e^+ - L_e^-|.$$

Thus the minimum in $\mathcal{U}_{n,c,d}$ is the minimum of the displayed expression over the finite set of rooted-tree systems satisfying the order and diameter constraints.

6 Prescribed diameter

Let

$$\mathcal{U}_{n,d} = \{G : G \text{ is a connected unicyclic graph of order } n \text{ and } \text{diam}(G) = d\}.$$

The feasible parameters are $n = 3, d = 1$, or $n \geq 4$ and $2 \leq d \leq n - 2$.

Theorem 6.1 (Maximum for prescribed diameter). *Assume $\mathcal{U}_{n,d} \neq \emptyset$. Then*

$$\max_{G \in \mathcal{U}_{n,d}} \text{Mo}_t(G) = \begin{cases} 0, & n = 3, d = 1, \\ 3, & n = 4, d = 2, \\ (n-3)^2, & d = 2, n \geq 5, \\ (n-4)(n-2), & d = 3, n \geq 6, \\ \max\{(n-d-1)(n-2), n(n-d-2)\}, & 4 \leq d \leq n-3, \\ n, & d = n-2, n \geq 5. \end{cases}$$

For $d = 2$ and $n \geq 5$, the extremal graph is S_n^+ . For $d \geq 3$, the extremal graphs are obtained from C_4 : for $d = 3$ all $n - 4$ non-cycle vertices are leaves attached to one cycle vertex; for $4 \leq d \leq n - 3$ the extremal type is the one-sided or two-sided type from Section 5; for $d = n - 2$ the extremal graph is C_4 with a pendent path of length $n - 4$ attached to one cycle vertex.

Proof. The diameter-two cases are immediate: S_4^+ has value 3, and for $n \geq 5$ the unique extremal construction is the triangle with all $n - 3$ leaves attached to one vertex, giving $(n - 3)^2$. For $d \geq 3$, maximize the formula of Theorem 5.2 over all feasible cycle lengths. Direct substitution shows that the optimum is obtained for $c = 4$. For $d = 3$, this yields $(n - 4)(n - 2)$. For $4 \leq d \leq n - 3$, we have $r = 2, m = n - 4$ and $K = n - d$, so the maximum is

$$\max\{(K - 1)(n - 2), n(K - 2)\} = \max\{(n - d - 1)(n - 2), n(n - d - 2)\}.$$

For $d = n - 2$, the $c = 4$ maximal-diameter boundary case gives value n . No other feasible cycle length gives a larger value. $\square$

Remark 6.2. For $4 \leq d \leq n - 3$, the one-sided construction gives the term $(n - d - 1)(n - 2)$ and the two-sided strongly unbalanced construction gives $n(n - d - 2)$. The former is larger exactly when $2(n - d) \leq n + 2$, that is, $d \geq (n - 2)/2$; at equality both constructions are extremal.

We next record the main minimum phenomena for prescribed diameter.

Theorem 6.3 (Zero criterion). *Assume $\mathcal{U}_{n,d} \neq \emptyset$. Then*

$$\min_{G \in \mathcal{U}_{n,d}} \text{Mo}_t(G) = 0$$

if and only if one of the following holds:

- $n = 2s$ is even and $d \geq s$;
- $n = 2s + 1$ is odd and either $d = s$ or $d \geq s + 2$.

Equivalently, for odd order the only missing critical value is $d = (n + 1)/2$.

Proof. If $d = \lfloor n/2 \rfloor$, the cycle C_n belongs to $\mathcal{U}_{n,d}$ and has terminal Mostar index zero. Non-cycle zero graphs must be even cycles with two opposite pendent paths. Let the path lengths be $p, q \geq 1$ and the cycle be C_{2r} . Then

$$n = 2r + p + q, \quad d = p + q + r = n - r.$$

Thus $r = n - d$ and $p + q = 2d - n$. Since $p, q \geq 1$, we need $2d - n \geq 2$. For $n = 2s$, this gives $d \geq s + 1$, together with the cycle case $d = s$. For $n = 2s + 1$, it gives $d \geq s + 2$, together with the cycle case $d = s$. $\square$

Theorem 6.4 (Odd-order critical diameter). *If $n = 2s + 1$ and $d = s + 1 = (n + 1)/2$, then*

$$\min_{G \in \mathcal{U}_{n,d}} \text{Mo}_t(G) = 2.$$

An extremal graph is obtained from the odd cycle $C_{n-2} = C_{2s-1}$ by attaching one leaf to each of two cycle vertices at distance $s - 1$.

Proof. The two leaves in the stated graph are at distance $1 + (s - 1) + 1 = s + 1 = d$. The two leaf edges contribute zero and the cycle contribution is 2 by the two-leaf cycle lemma. The zero criterion excludes value zero. Graphs with one leaf have value at least 3, and graphs with at least three leaves have leaf-edge contribution at least 3. Thus value 2 is best possible. $\square$

Proposition 6.5 (Diameter two). *For $n = 4$, the minimum in $\mathcal{U}_{4,2}$ is 0, attained by C_4 . For $n = 5$, the minimum in $\mathcal{U}_{5,2}$ is 0, attained by C_5 . If $n \geq 6$ and $d = 2$, then the only graph is S_n^+ , and hence*

$$\min_{G \in \mathcal{U}_{n,2}} \text{Mo}_t(G) = \max_{G \in \mathcal{U}_{n,2}} \text{Mo}_t(G) = (n - 3)^2.$$

Proof. The only cycles of diameter two are C_4 and C_5 . If a unicyclic graph of diameter two has a leaf, its cycle must be a triangle; otherwise a leaf is at distance at least three from some cycle vertex. On a triangle, leaves attached to two different cycle vertices would be at distance three, so all leaves must be attached to one vertex. This gives S_n^+ . $\square$

Remark 6.6 (Finite optimization for small diameters). For general $d < \lfloor n/2 \rfloor$, the order is too large relative to the diameter for a cycle or a two-leaf graph to realize the minimum. Extremal graphs usually need at least three leaves distributed as evenly as possible. The finite optimization of Section 5 gives an exact procedure: enumerate feasible cycle lengths and rooted-tree systems, and minimize

$$\sum_{e \notin E(C)} |k - 2s_e| + \sum_{e \in E(C)} |L_e^+ - L_e^-|.$$

Thus the minimum problem in the small-diameter range is a balanced distribution problem for leaves.

7 Concluding remarks and further problems

We extended the terminal Mostar index from trees to unicyclic graphs and obtained the following results. First, among all n -vertex unicyclic graphs, the minimum is always zero, attained only by cycles and even-cycle opposite double pendent paths; for $n \geq 6$, the maximum is $(n - 3)^2$ and is attained uniquely by S_n^+ . This differs from the ordinary Mostar index and shows the effect of terminalization.

Second, for the class with prescribed cycle length c , both the minimum and the maximum have simple closed formulae. The minimum reflects the principle that two leaves should be placed as symmetrically as possible on the cycle, while the maximum reflects concentration of all leaves at a single cycle vertex.

Third, for prescribed cycle length and prescribed diameter, the maximum can still be determined explicitly. In the intermediate range, two extremal mechanisms compete: a one-sided concentrated construction and a two-sided strongly unbalanced construction. The minimum is fully characterized for zero cases, for the odd-cycle maximal-diameter critical case, and within the two-leaf subfamily; the remaining cases reduce to a finite balanced optimization problem.

Fourth, for prescribed diameter alone, maximizing over the cycle length yields an explicit formula. In particular, for $4 \leq d \leq n - 3$ the maximum is

$$\max\{(n - d - 1)(n - 2), n(n - d - 2)\}.$$

Several problems remain natural:

- (1) Find closed formulae for the minimum in the small-diameter range $d < \lfloor n/2 \rfloor$.

- (2) Study terminal Mostar extremal problems for bicyclic graphs, cactus graphs and chemical graphs.
- (3) Investigate weighted terminal Mostar indices, edge-terminal Mostar indices and the corresponding polynomials.
- (4) Develop efficient algorithms for the finite balanced optimization problem with prescribed n, c, d .

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